

Calc 3 – Gradient & Directional Derivatives

Worksheet A

1. Find ∇f for $f(x, y) = x^2y + y^3$.
2. Find the directional derivative of $f(x, y) = x^2 - y^2$ at $(1, 2)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.
3. Find the direction of maximum increase of $f(x, y, z) = xyz$ at $(1, 2, 3)$ and the magnitude of that maximum rate.
4. Find the tangent plane to the level surface $x^2 + y^2 - z^2 = 1$ at $(1, 1, 1)$.
5. For $f(x, y, z) = xe^{yz}$, compute $D_{\mathbf{u}}f(1, 0, 2)$ where $\mathbf{u}$ is the unit vector toward $(2, 1, 0)$.

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Worksheet B

1. Find ∇f for $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$.
2. Find the directional derivative of $f(x, y) = \ln(x^2 + y^2)$ at $(1, 2)$ toward $(0, 0)$.
3. At which point on the surface $x^2 + 2y^2 + 3z^2 = 6$ is the tangent plane parallel to the xy -plane?
4. Find the rate of change of $f(x, y) = xy + y^2$ at $(2, 3)$ in the direction making angle $\pi/3$ with positive x -axis.
5. Find the equation of the normal line to the surface $z = x^2 + y^2$ at the point $(1, 2, 5)$.

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Answer Key (Worksheets A & B)

Worksheet A.

1. $\nabla f = \langle 2xy, x^2 + 3y^2 \rangle$.
2. $\nabla f = \langle 2x, -2y \rangle$, at $(1, 2)$: $\langle 2, -4 \rangle$. $\hat{\mathbf{v}} = \frac{1}{5}\langle 3, 4 \rangle$. $D_{\mathbf{u}}f = \frac{1}{5}(6 - 16) = -2$.
3. $\nabla f = \langle yz, xz, xy \rangle$, at $(1, 2, 3)$: $\langle 6, 3, 2 \rangle$. Max increase direction: $\langle 6, 3, 2 \rangle / \sqrt{49}$. Max rate = $\sqrt{49} = 7$.
4. $\nabla F = \langle 2x, 2y, -2z \rangle$, at $(1, 1, 1)$: $\langle 2, 2, -2 \rangle$. Plane: $2(x - 1) + 2(y - 1) - 2(z - 1) = 0$ i.e. $x + y - z = 1$.
5. $\nabla f = \langle e^{yz}, xze^{yz}, xye^{yz} \rangle$, at $(1, 0, 2)$: $\langle 1, 2, 0 \rangle$. $\mathbf{u} = \frac{(2-1, 1-0, 0-2)}{\sqrt{1+1+4}} = \frac{\langle 1, 1, -2 \rangle}{\sqrt{6}}$. $D_{\mathbf{u}}f = \frac{1}{\sqrt{6}}(1 + 2 + 0) = 3/\sqrt{6} = \sqrt{6}/2$.

Worksheet B.

1. Let $r = \sqrt{x^2 + y^2 + z^2}$. Then $\nabla f = \frac{\langle x, y, z \rangle}{\sqrt{x^2 + y^2 + z^2}}$.
2. $\nabla f = \frac{\langle 2x, 2y \rangle}{x^2 + y^2}$, at $(1, 2)$: $\frac{1}{5}\langle 2, 4 \rangle$. Direction toward origin: $\frac{(0-1, 0-2)}{\sqrt{5}} = \frac{1}{\sqrt{5}}\langle -1, -2 \rangle$. $D = \frac{1}{5\sqrt{5}}(-2 - 8) = -\frac{2}{\sqrt{5}}$.
3. Tangent plane parallel to xy -plane $\Leftrightarrow$ normal vertical $\Leftrightarrow \nabla F = \langle 2x, 4y, 6z \rangle$ has x -and- y components zero $\Rightarrow x = 0, y = 0$. Then $3z^2 = 6 \Rightarrow z = \pm\sqrt{2}$. Points: $(0, 0, \pm\sqrt{2})$.
4. $\nabla f = \langle y, x + 2y \rangle$, at $(2, 3)$: $\langle 3, 8 \rangle$. $\mathbf{u} = \langle \cos(\pi/3), \sin(\pi/3) \rangle = \langle 1/2, \sqrt{3}/2 \rangle$. $D = 3/2 + 4\sqrt{3} = 3/2 + 4\sqrt{3}$.
5. ∇F of $F = z - x^2 - y^2$ at $(1, 2, 5)$: $\langle -2, -4, 1 \rangle$. Normal line: $(x, y, z) = (1, 2, 5) + t\langle -2, -4, 1 \rangle$.